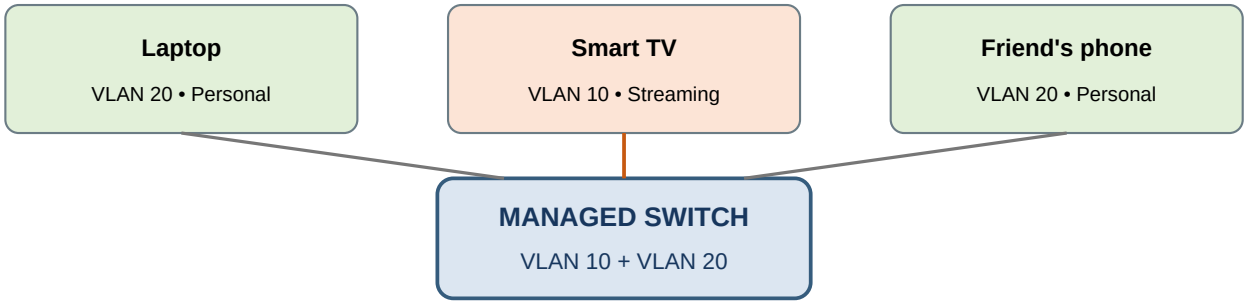
VLAN Configuration and Switching Practical

Question 1. Draw a simple network diagram showing three devices connected to a switch, and label the VLAN for each device.

Answer: A VLAN (Virtual Local Area Network) logically separates devices on the same physical switch into different Layer 2 networks. In this example, the Smart TV is placed in VLAN 10 for streaming, while the laptop and friend's phone are placed in VLAN 20 for personal devices.



VLAN 10: streaming devices | VLAN 20: personal devices

Devices in different VLANs need a router / Layer 3 switch to communicate.

Example diagram for the assignment (ports and VLAN IDs are illustrative).

Explanation: Each switch port is assigned to one access VLAN. VLAN 10 and VLAN 20 are separate broadcast domains, so devices in different VLANs normally cannot communicate directly through a Layer 2 switch. Communication between VLANs requires inter-VLAN routing through a router or a Layer 3 switch, with suitable IP addressing and routing rules.

Important terms

VLAN ID: the number identifying a VLAN. **Access port:** a switch port that normally carries traffic for one assigned VLAN. **Inter-VLAN routing:** Layer 3 forwarding between different VLANs.

Question 2. In Packet Tracer or NetSim, create VLAN 10 and VLAN 20, assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Answer: The following example uses a Cisco 2960-style switch in Packet Tracer. Use five PCs connected to switch ports Fa0/1 through Fa0/5. The exact port names may differ if your selected switch model uses another interface numbering scheme.

Step 1 — Build the topology

Place one 2960 switch and five PCs. Connect each PC's FastEthernet port to a switch access port using copper straight-through cables (or use Packet Tracer's automatic connection option). Open the switch CLI and press Enter.

Step 2 — Create the VLANs

```
enable
configure terminal
vlan 10
 name STREAMING
exit
vlan 20
 name PERSONAL
exit
```

Step 3 — Assign three ports to VLAN 10 and two ports to VLAN 20

```
interface range fastEthernet 0/1 - 3
 switchport mode access
 switchport access vlan 10
exit
interface range fastEthernet 0/4 - 5
 switchport mode access
 switchport access vlan 20
exit
end
write memory
```

The first interface range assigns Fa0/1, Fa0/2, and Fa0/3 to VLAN 10. The second assigns Fa0/4 and Fa0/5 to VLAN 20. The `switchport mode access` command makes these ports access ports rather than trunk ports.

Step 4 — Set example IPv4 addresses on the PCs

PC	Switch port	VLAN	Example IPv4 address	Subnet mask
PC0	Fa0/1	10	192.168.10.11	255.255.255.0
PC1	Fa0/2	10	192.168.10.12	255.255.255.0
PC2	Fa0/3	10	192.168.10.13	255.255.255.0
PC3	Fa0/4	20	192.168.20.11	255.255.255.0
PC4	Fa0/5	20	192.168.20.12	255.255.255.0

In Packet Tracer, click each PC → Desktop → IP Configuration, then enter the IP address and subnet mask from the table. Leave Default Gateway blank for this Layer 2-only test. Do not give PCs in VLAN 10 the VLAN 20 subnet, or vice versa.

Question 2 — Verification and expected results

Step 5 — Verify the VLAN and port assignments on the switch

```
enable
show vlan brief
show interfaces status
```

In `show vlan brief`, check that Fa0/1–Fa0/3 appear under VLAN 10 and Fa0/4–Fa0/5 appear under VLAN 20. If a port is missing from the expected VLAN, re-enter the interface configuration for that port.

Step 6 — Test connectivity

On a PC, open Desktop → Command Prompt. Run the following tests:

```
PC0> ping 192.168.10.12
Expected: success (PC0 to PC1, both VLAN 10)

PC0> ping 192.168.20.11
Expected: failure (PC0 VLAN 10 to PC3 VLAN 20)

PC3> ping 192.168.20.12
Expected: success (PC3 to PC4, both VLAN 20)
```

Why does the cross-VLAN ping fail? A Layer 2 switch does not route packets between VLANs. In this topology there is no router or Layer 3 switch configured to route between 192.168.10.0/24 and 192.168.20.0/24. A failed cross-VLAN ping is therefore expected. If inter-VLAN routing is later configured, the result may change.

Troubleshooting checklist

• If same-VLAN ping fails, check the PC IP address and mask, cable/link lights, switch port status, and VLAN assignment.
• If `show vlan brief` lists the port in the wrong VLAN, configure the correct `switchport access vlan` value.
• Wait a few seconds after changing a port, then retry the ping.
• A PC in another VLAN will not become reachable merely by changing its switch port if its IP address remains in the old subnet.

Question 3. Move a PC from VLAN 10 to VLAN 20. Show the before-and-after configuration commands and test connectivity.

Answer: Assume PC2 is connected to switch port Fa0/3 and is initially in VLAN 10. A second test PC, PC4, is in VLAN 20 with IP address 192.168.20.12.

Before — confirm the current VLAN and configure the port for VLAN 10

```
enable
show vlan brief
configure terminal
interface fastEthernet 0/3
 switchport mode access
 switchport access vlan 10
end
show vlan brief
```

Configure PC2 initially with IP address 192.168.10.13 and subnet mask 255.255.255.0. Before moving it, it should be able to ping another VLAN 10 PC, for example 192.168.10.11, assuming the link and addressing are correct.

Move — change the access VLAN on the same switch port

```
enable
configure terminal
interface fastEthernet 0/3
 switchport access vlan 20
end
show vlan brief
```

After the command, Fa0/3 should appear under VLAN 20 in `show vlan brief`. This command changes the Layer 2 VLAN assignment of the switch port; it does not automatically change the PC's IP address.

After — update the PC's IP address and test again

If no DHCP server is available, open PC2 → Desktop → IP Configuration and change the address to 192.168.20.13 with mask 255.255.255.0. Keep the IP unique; do not reuse PC4's 192.168.20.12 address.

```
PC2> ping 192.168.20.12
Expected: success (PC2 and PC4 are now both in VLAN 20)

PC2> ping 192.168.10.11
Expected: failure in this Layer 2-only topology
```

Result: The PC's switch port has moved from VLAN 10 to VLAN 20. To communicate correctly in the new VLAN, the PC also needs an IP address belonging to VLAN 20's subnet. If the lab includes DHCP, renew or request an address from the VLAN 20 DHCP scope instead of setting a static address.

Question 4. Write a command list or script to automate assigning 10 switch ports to VLAN 30, using a loop or range command if supported.

Answer: Cisco IOS supports `interface range`, which lets the administrator configure several interfaces together without repeating the same commands for every port. This example assigns FastEthernet ports Fa0/1 through Fa0/10 to VLAN 30.

Step 1 — Create VLAN 30

```
enable
configure terminal
vlan 30
 name VLAN30_USERS
exit
```

Step 2 — Apply the VLAN to all ten ports with one interface-range command

```
interface range fastEthernet 0/1 - 10
 switchport mode access
 switchport access vlan 30
exit
end
write memory
```

The range `fastEthernet 0/1 - 10` selects Fa0/1 through Fa0/10 on common Packet Tracer Cisco switch models. If your device uses GigabitEthernet ports, use the matching interface type and range, such as `interface range gigabitEthernet 0/1 - 10`, only if those ports actually exist on your switch.

Step 3 — Verify the result

```
show vlan brief
show interfaces status
```

Check that VLAN 30 exists and that all ten intended ports are listed in VLAN 30. Save the configuration with `write memory` or `copy running-config startup-config`.

Optional Python-style pseudocode (for a simulator/API that supports scripted configuration)

```
# Pseudocode only: not a command to paste into the Cisco IOS CLI
for port in range(1, 11):
    configure_interface(f"FastEthernet0/{port}")
    set_switchport_mode("access")
    set_access_vlan(30)
```

Use the Cisco IOS `interface range` version in Packet Tracer's switch CLI. The Python-style loop is illustrative pseudocode and requires a simulator or automation library that exposes switch configuration functions.

Final practical checklist

1. Create the required VLANs before assigning ports.
2. Confirm port-to-VLAN membership using `show vlan brief`.
3. Use different IP subnets for the separate VLANs in this lab.
4. Test same-VLAN connectivity and cross-VLAN isolation.
5. When moving a PC, update its IP configuration or renew DHCP as appropriate.
6. Save the switch configuration after verification.